

GI IMAGING (Radiology)

PLAIN ABDOMINAL X-RAY

- Standard = **supine AP**. Free intraperitoneal air best on **ERECT CHEST X-ray**. Can't sit → **lateral decubitus**.
- **Gas pattern**: large amounts normal in stomach + colon. Stomach = LUQ (gastric rugae bands). Small bowel = little gas.

SBO vs LBO (HIGH-YIELD)

| Feature | Small bowel obstruction | Large bowel obstruction |

|---|---|---|

| Location | **Central** ("frame" of colon) | **Peripheral** (transverse/sigmoid central) |

| Loops | Numerous | Few |

| Mucosal markings | **Valvulae conniventes** (cross full width, "stack of coins") | **Haustra** (incomplete bands, absent distal to splenic flexure) |

| Curve | Tight | Wide |

| Diameter | **3–6 cm** | **>6 cm** |

- **Mechanical obstruction causes**: lumen (gallstone ileus), wall (tumour, Crohn's), extrinsic (adhesions, hernia). Non-mechanical dilatation: paralytic ileus, ischaemia, IBD.
- **CT**: shows **transition point** (dilated→collapsed), confirms/excludes mass; **small bowel feces sign**.

Pneumoperitoneum

- Commonest spontaneous cause = **perforated peptic ulcer** (2/3 recognizable radiologically). Normal post-laparotomy (absorbs <7 days adults, <24h children). **Erect CXR: curvilinear gas under right hemidiaphragm** (between diaphragm + liver).

Abdominal calcification

- **Appendicolith** + appendicitis features → strong indication (often gangrene/perforation). **Phleboliths** (pelvic veins, mistaken for stones). Calcified mesenteric LN (old TB, mobile). **Vascular = curvilinear (AAA wall)**. Uterine fibroids. **Dermoid cyst = tooth shape**. Adrenal (haemorrhage/TB/Addison's minority). **Gallstones 20% visible (dense rim, lucent centre)**. **Pancreatic calcification = chronic pancreatitis (L1 level)**. Renal **staghorn stones**.

GI IMAGING MODALITIES

- **Barium sulphate**: best contrast/opacification. **Contraindicated if perforation suspected** (causes severe peritonitis if leak).
- **Gastrografin (water-soluble)**: for **suspected perforation/anastomotic leak**; less radio-opaque; **irritant to lungs** → **pulmonary oedema** if aspirated.
- **Double-contrast** (barium + air): mucosal detail (stomach, colon).
- **Ultrasound**: fluid, bowel wall; **pyloric stenosis, intussusception, appendicitis, diverticulitis**.
- **EUS: depth of tumour invasion** (esophagus, gastric, rectal); small pancreatic/duodenal tumours; guide FNA/biopsy.
- **CT**: full GI wall + surrounding fat (not mucosa); staging tumours, inflammation, obstruction, trauma. Mesenteric CTA for ischaemia.
- **MRI**: limited (peristalsis artefact) but **rectal carcinoma staging + MR enterography (small bowel/Crohn's, antiperistaltic agent)**.
- **FDG-PET/CT**: metastatic disease in GI cancers.

Descriptive terms (HIGH-YIELD)

- **Ulcer:** breach of mucosa, barium fills crater (projects beyond wall).
- **Filling defect:** intraluminal (food, surrounded by barium), intramural (tumour/leiomyoma, indents), extramural (compresses, mucosa preserved).
- **Stricture:** circumferential narrowing. **Benign = tapering ends; malignant = abrupt with overhanging edges ("shouldering").**
- **Fat stranding** (CT/MRI) / echogenic fat (US) = inflammation. Wall thickening. FDG uptake (PET).

ESOPHAGUS

- Normal: smooth, indentations from aortic arch + left atrium. **Tertiary contractions = corkscrew oesophagus.**
- **Strictures:**
- **Carcinoma:** irregular lumen + **shouldered edges**, several cm; CT/EUS/PET for staging.
- **Peptic:** lower end, **short, smooth, tapering.**
- **Achalasia:** failure of cardiac sphincter relaxation → **smooth tapered narrowing at lower end + dilated oesophagus above with absent peristalsis** ("bird's beak"); plain film = food residue, aspiration → consolidation/bronchiectasis.
- **Corrosive:** long stricture from aortic arch level; smooth/tapered ± irregular.
- **Leiomyoma** (benign): smooth rounded indentation.
- **Dilatation:** with obstruction (cancer/achalasia); **without obstruction = scleroderma** (lower 2/3 smooth muscle).
- **Zenker's diverticulum:** posterior pharynx (inferior constrictor weakness). **Web:** cervical, anterior, **Plummer-Vinson.**
- **Perforation:** X-ray pneumomediastinum + effusion; **water-soluble contrast** shows leak; **CT with oral contrast most sensitive.**

STOMACH & DUODENUM

- **Peptic ulcer:** gastric = benign or malignant (**always scope**); duodenal = almost always benign. Ulcer = barium projection beyond mucosa; duodenal cap deformed by scarring.

Benign vs malignant gastric ulcer (HIGH-YIELD)

| Benign | Malignant |

|---|---|

| Crater **projects beyond** gastric contour | Does NOT protrude beyond contour (within outline) |

| Smooth, rounded, deep crater | Irregular, shallow crater |

| Smooth folds **reaching** ulcer margin | Nodular folds **NOT reaching** margin |

| **Hampton's line** | **Carman meniscus sign** |

- **Gastric carcinoma:** irregular filling defect; or focal wall thickening; or **linitis plastica** (diffuse thickening, rigid stomach, loss of folds). Staging: CT/EUS/laparoscopy.
- **GIST:** smooth round submucosal filling defect; usually benign; **60–70% in stomach.**
- **Lymphoma:** stomach = most frequent GI site; diffuse wall thickening + bulky lymphadenopathy.
- **Hiatus hernia:** **sliding** (GEJ + stomach above diaphragm, incompetent sphincter → reflux/stricture); **rolling/paraesophageal** (fundus herniates, GEJ competent). CXR = retrocardiac mass + air-fluid level.

SMALL & LARGE INTESTINE

Techniques

- **Small bowel follow-through** (drink barium): Crohn's, stricture, malabsorption. **Enteroclysis** (NJ tube + barium + methylcellulose): best mucosal detail. **Barium enema** (double contrast): colon.
- CT, US (bowel thickening, appendix, intussusception), MRI (rectal/anal, MR enterography). **Tc-99m for Meckel's**; octreotide/MIBG for carcinoid.

Dilatation/narrowing

- Small >3cm (malabsorption/ileus/SBO); large >6cm (obstruction, ileus, volvulus, **toxic megacolon**, Hirschsprung).
- **Colonic stricture site**: diverticular = **sigmoid**; ischaemic = **splenic flexure–sigmoid**; Crohn's/TB = **caecum**. Neoplastic = shouldered, irregular, <6cm; benign = tapered.

IBD (HIGH-YIELD)

| Feature | Ulcerative colitis | Crohn's |

|---|---|---|

| Distribution | **Continuous from rectum** | **Skip lesions**, terminal ileum |

| Ulcers | Shallow (deep if severe) | Deep → **cobblestone** |

| Haustra | **Lost** → rigid tube ("lead pipe") | — |

| Other | **Pseudopolyps**, backwash ileitis | **String sign**, fistulae, bowel separation, caecal contraction |

| CT/MRE | Wall thickening, "**target sign**" (fatty submucosa) | Thick wall + **fat stranding**, prominent mesenteric vessels |

| Complication | **Toxic megacolon**; stricture rare (→ think cancer) | Strictures, fistulae, abscess |

Diverticular disease

- Sac-like mucosal outpouchings through muscle; commonest **sigmoid**. **Diverticulitis** = infected (thick wall + fat stranding ± perforation/abscess). Stricture **indistinguishable from carcinoma**.

Appendicitis

- Clinical mostly. US/CT: **distended >6mm, non-compressible appendix, thick wall, free fluid, fat inflammation**; abscess = RIF mass.

Obstruction & emergencies

- **SBO = adhesions/hernia; LBO = tumours**. CT transition zone; adhesions can't be directly imaged.
- **Volvulus**: bowel twists on mesentery; **sigmoid commonest** (then caecum). **Coffee-bean sign** (AXR); **whirl sign** (CT, twisted vessels).
- **Intussusception**: invagination. **Children = idiopathic** (US target/doughnut, air/CO2 enema reduction); **adults = neoplasm lead point** (CT sausage-shaped mass with mesenteric fat).
- **Mesenteric infarction**: SMA occlusion → wall thickening, **gas in bowel wall**; CT angiography shows thrombus. **Ischaemic colitis**: splenic flexure–sigmoid; **thumbprint sign** (mucosal oedema/haemorrhage).
- **TB**: contracted caecum, dilated terminal ileum, peritoneal nodules + necrotic nodes — mimics Crohn's/malignancy → biopsy.

Polyps & tumours

- **Polyp malignancy risk**: >2cm, short thick stalk, irregular surface. <1cm rarely cancer.
- **Carcinoma**: commonest **rectosigmoid + caecum**. Rectosigmoid → annular stricture + obstruction; **caecal** → **large without obstruction (anaemia + weight loss)**. Barium = **shouldered irregular stricture <6cm** or fungating mass. **MRI for rectal staging**; CT for mets; PET before curative surgery.

- **Lymphoma:** marked wall thickening **WITHOUT lumen narrowing** (differentiates from Crohn's); mesenteric/para-aortic nodes.

HIGH-YIELD ONE-LINERS

- **SBO = central, valvulae conniventes (stack of coins), 3–6cm; LBO = peripheral, haustra, >6cm.**
- **Free air = erect CXR (perforated PU commonest).** Barium contraindicated if perforation → use Gastrografin.
- **Benign ulcer projects beyond contour (Hampton's line); malignant within contour (Carman sign).**
- **Achalasia = bird's beak; scleroderma = dilated no obstruction; corrosive = long from aortic arch.**
- **UC = continuous, lost haustra (lead pipe), pseudopolyps, toxic megacolon; Crohn's = skip lesions, cobblestone, string sign, fistulae.**
- **Volvulus = coffee-bean / whirl sign; sigmoid commonest.**
- **Intussusception: children idiopathic (reduce by enema); adults = tumour lead point.**
- **Ischaemic colitis = thumbprint sign, splenic flexure–sigmoid.**
- **Lymphoma = thick wall, NO narrowing (vs Crohn's). Polyp malignant if >2cm.**
- **Linitis plastica = rigid stomach; GIST = submucosal; gastric lymphoma commonest GI lymphoma site.**

GI IMAGING (Radiology) — Revision Table

Bold = high-yield. Plain film + modalities + esophagus/stomach + intestine.

PLAIN FILM & MODALITIES	
Topic	Key points
Plain AXR	Supine AP standard; free air = ERECT CXR ; can't sit → lateral decubitus
SBO vs LBO	SBO: central, valvulae conniventes (stack of coins, cross full width), 3–6cm, numerous. LBO: peripheral, haustra (incomplete bands), >6cm, few
Pneumoperitoneum	Commonest = perforated PU ; curvilinear gas under R hemidiaphragm on erect CXR
Calcifications	Appendicolith (appendicitis), phlebolith, gallstone 20% (dense rim), pancreatic = chronic pancreatitis (L1) , dermoid = tooth, AAA = curvilinear, staghorn = renal
Contrast	Barium = best detail but CONTRAINDICATED if perforation ; Gastrografin = perforation/leak (aspirate → pulmonary oedema)
Modalities	EUS = tumour depth ; MRI = rectal cancer staging + MR enterography (Crohn's) ; CT = wall/staging/obstruction; PET = mets
Descriptive	Benign stricture = tapered ; malignant = shouldering (abrupt overhanging). Fat stranding = inflammation
ESOPHAGUS	
Condition	Imaging
Carcinoma	Irregular lumen + shouldered edges ; EUS/CT/PET staging
Peptic stricture	Lower end, short, smooth, tapered
Achalasia	Smooth tapered lower narrowing + dilated above, absent peristalsis (bird's beak) ; aspiration
Corrosive	Long stricture from aortic arch level
Scleroderma	Dilatation WITHOUT obstruction (lower 2/3)
Zenker / web	Zenker = posterior pharynx; web = cervical + Plummer-Vinson
Perforation	Pneumomediastinum; water-soluble contrast / CT oral contrast
STOMACH	
Condition	Imaging
Benign ulcer	Projects BEYOND contour, smooth deep crater, folds reach margin, Hampton's line
Malignant ulcer	Within contour, irregular shallow, folds DON'T reach, Carman meniscus sign
Gastric carcinoma	Irregular filling defect / focal thickening / linitis plastica (rigid stomach, lost folds)
GIST	Smooth round submucosal filling defect; 60–70% stomach
Lymphoma	Stomach = commonest GI site ; diffuse thickening + bulky nodes
Hiatus hernia	Sliding (GEJ above, reflux); rolling (fundus, GEJ competent); CXR retrocardiac air-fluid level
SMALL & LARGE INTESTINE	
Condition	Imaging
UC	Continuous from rectum, lost haustra (lead pipe), shallow ulcers, pseudopolyps, target sign, toxic megacolon ; stricture rare → cancer
Crohn's	Skip lesions, terminal ileum, deep ulcers → cobblestone, string sign, fistulae , fat stranding
Diverticular	Sigmoid commonest; diverticulitis = thick wall + fat stranding; stricture mimics cancer
Appendicitis	US/CT: >6mm non-compressible, thick wall, fat inflammation ; abscess = RIF mass
Volvulus	Sigmoid commonest; coffee-bean sign (AXR), whirl sign (CT)
Intussusception	Children = idiopathic (target, reduce by air/CO2 enema) ; adults = tumour lead point (CT sausage mass)
Mesenteric ischaemia	SMA occlusion, gas in wall, CTA thrombus. Ischaemic colitis = thumbprint sign (splenic flexure–sigmoid)
TB	Contracted caecum + dilated ileum; mimics Crohn's → biopsy
Carcinoma	Rectosigmoid (annular stricture, obstruct) + caecal (large, no obstruction, anaemia/weight loss) ; shouldered <6cm ; MRI rectal staging

Condition	Imaging
Lymphoma	Marked wall thickening WITHOUT narrowing (vs Crohn's)
Polyps	Malignant if >2cm, thick short stalk, irregular